

Background and Motivation

- In the U.S., more than 1 out of 3 older adults aged ≥ 65 years fall each year [1].
- Successful balance recovery at the onset of a fall requires quick & strong lower limb reflex responses, involving high rate of torque development (RTD) [2].
- Age-related changes in muscle contractile properties result in a decline in RTD during fast voluntary contractions.
- This research aims to assist lower limb balance recovery response in older adults at the onset of a fall, by simultaneously providing assistive torque at joint level and Functional Electrical Stimulation (FES) at muscular level.

Objectives

1. Develop a lower limb hybrid FES-exoskeleton to assist balance recovery at fall onset- integrating FES with robotic actuators
2. Evaluate the efficacy of the proposed system in assisting the user to recover balance against unexpected slip/trip-like perturbations in controlled environment.

Research Questions

1. Which biomechanical features indicative of a fall can accurately detect at its early onset to provide the longest pre-impact lead time with high reliability?
2. Can a hybrid FES-exoskeleton system arrest a fall in real-time by augmenting biomechanical responses in targeted lower limb joints and muscle groups?

Aims and Methods

Aim 1: Design and fabrication of a balance perturbation system to induce slip/trip-like perturbations during treadmill walking

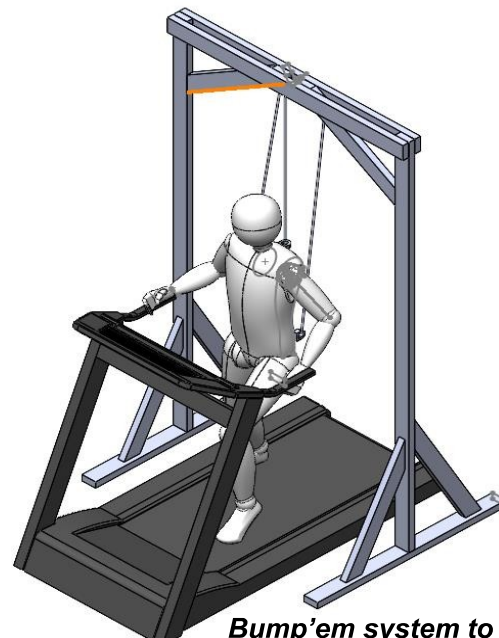
- Method: Implement a bump-emulation system (i.e., bump'e,m) onto a treadmill, in conjunction to a truss with safety harness, to simulate gait perturbation during walking and record data via force sensor and IMU.

Aim 2: Developing a pre-impact fall detection algorithm to detect loss of balance after unexpected slip/trip-like perturbations using wearable sensors (IMU & pressure insoles)

- Method: Selection of fall indicator parameters, identifying number of sensors and optimal sensor placement position, selection of classifier method and validation

Aim 3: Assisting/augmenting balance recovery response to restore gait/postural stability after pre-impact fall detection using a lower limb hybrid FES-exoskeleton system

- Method: Design and fabrication of the wearable robotic device and developing a hybrid control algorithm combining feedback control for robotic actuator and feedforward control for FES.



Bump'em system to induce unexpected slip/trip-like perturbations

Significance and Broader Impact

- Successful implementation will result in reducing fall related injuries & deaths, enhancing healthy longevity, and improving quality of life among elderly
- Once validated, it may reduce financial burden of health care sector by minimizing the medical and rehabilitation expenditure associated with fall

Progress and Outcomes

In progress:

- CAD model of gait perturbation system
- Truss system construction around treadmill
- Finish assembling by March 2023
- IRB Protocol for human subject testing

Control Block Diagram for Hybrid FES-Exoskeleton

